# Kinematics

## Section 1: Constant Acceleration & SUVAT Equations

**1** A car accelerates uniformly from rest to a speed of 24 m/s over a distance of 144 m. Find its acceleration.

**a.** $2\ \text{m/s}^2$

**b.** $1.5\ \text{m/s}^2$

**c.** $3\ \text{m/s}^2$

**d.** $4\ \text{m/s}^2$

**2** A ball is thrown vertically upwards with an initial speed of 20 m/s from ground level. Taking $g = 10\ \text{m/s}^2$, what is the maximum height reached?

**a.** 20 m

**b.** 40 m

**c.** 15 m

**d.** 25 m

**3** A train decelerates uniformly from 30 m/s to 10 m/s in 8 seconds. How far does the train travel during these 8 seconds?

**a.** 160 m

**b.** 140 m

**c.** 180 m

**d.** 120 m

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## Section 2: Velocity-Time & Displacement-Time Graphs

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**4** On a velocity-time graph, what physical quantities are represented by the gradient and the area under the graph, respectively?

**a.** Acceleration and Displacement

**b.** Displacement and Acceleration

**c.** Speed and Distance

**d.** Force and Momentum

**5** A particle moves according to the following $v$-$t$ graph: accelerates from rest to $15\ \text{m/s}$ in $6$ s, travels at constant $15\ \text{m/s}$ for $10$ s, then decelerates to rest in $4$ s. What is the total distance traveled?

**a.** $225$ m

**b.** $250$ m

**c.** $200$ m

**d.** $270$ m

**6** A particle's displacement is given by $s(t) = 4t - t^2$ for $0 \le t \le 4$. At what time does the particle momentarily come to rest?

**a.** $t = 2$ s

**b.** $t = 4$ s

**c.** $t = 1$ s

**d.** $t = 0$ s

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## Section 3: Variable Acceleration via Calculus

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**7** The velocity of a particle moving along a straight line is given by $v(t) = 6t^2 - 18t + 12$ for $t \geq 0$. Find the acceleration at $t = 3$.

**a.** $18\ \text{m/s}^2$

**b.** $12\ \text{m/s}^2$

**c.** $24\ \text{m/s}^2$

**d.** $6\ \text{m/s}^2$

**8** A particle starts from the origin at $t = 0$ with velocity $v(t) = 3t^2 - 4t$. Find its displacement $s(t)$ at $t = 3$.

**a.** $9\ \text{m}$

**b.** $18\ \text{m}$

**c.** $15\ \text{m}$

**d.** $6\ \text{m}$

**9** A particle has acceleration $a(t) = 4 - 2t\ \text{m/s}^2$. If $v(0) = 5\ \text{m/s}$, find the maximum velocity of the particle.

**a.** $9\ \text{m/s}$

**b.** $7\ \text{m/s}$

**c.** $8\ \text{m/s}$

**d.** $11\ \text{m/s}$

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## Section 4: Multi-Stage Motion & Exam Challenges

**10** A stone is dropped from the top of a cliff of height $H$. One second later, a second stone is thrown vertically downwards with speed $20\ \text{m/s}$. Both stones hit the sea at the same instant. Taking $g = 10\ \text{m/s}^2$, find $H$.

**a.** $45\ \text{m}$

**b.** $80\ \text{m}$

**c.** $60\ \text{m}$

**d.** $35\ \text{m}$

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**11** A police car initially at rest begins accelerating at $3\ \text{m/s}^2$ the moment a speeding car passes it at a constant speed of $24\ \text{m/s}$. How long does it take the police car to overtake the speeding car?

**a.** 16 s

**b.** 12 s

**c.** 8 s

**d.** 20 s

**12** A particle moving in a straight line passes through a fixed point $O$ with velocity $8\ \text{m/s}$. Its acceleration is $a = -0.5v\ \text{m/s}^2$. What distance does it travel before its speed halves to $4\ \text{m/s}$?

**a.** 8 m

**b.** 4 m

**c.** 12 m

**d.** 16 m

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